D. Y. Patil Agricultural and Technical University

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PROJECT REPORT ON

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

Department

Master Of Computer Applications

(MCA Sandwich Sem II)

Submitted By

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INTRODUCTION

The housing market plays a vital role in the economy, affecting both personal wealth and investment decisions. Understanding property prices and the factors that influence them is essential for buyers, sellers, and investors. Housing prices are determined by several factors such as location, property size, number of bedrooms, age of the property, and available amenities. Analysing housing data helps in identifying patterns, trends, and correlations that guide informed decisions in real estate.

This project, *"Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau"*, focuses on using data visualization to explore housing market trends. The primary objective is to examine how different property features affect sale prices. Tableau is used to create interactive dashboards and storyboards that present insights in a clear and engaging way, making complex data easier to understand.

Interactive visualizations allow users to detect trends, anomalies, and relationships in the data that are not immediately obvious from raw tables. By leveraging these tools, stakeholders can make better decisions regarding property investments, pricing strategies, and market analysis. Overall, this project demonstrates the power of data-driven visualization in understanding housing market dynamics effectively.

OBJECTIVES

The primary objective of this project is to analyse housing market trends and understand the factors that influence house sale prices. The project aims to provide clear insights into how various property features such as location, size, number of bedrooms, age, and available amenities affect pricing.

The specific objectives of the project are:

1. To analyse housing market patterns and identify trends in property prices.

2. To study the relationship between house features and their sale prices.

3. To use data visualization techniques to present complex housing data in an interactive and easy-to-understand format.

4. To develop interactive dashboards and storyboards in Tableau that allow users to explore the data dynamically.

5. To assist buyers, sellers, and investors in making informed decisions based on data-driven insights.

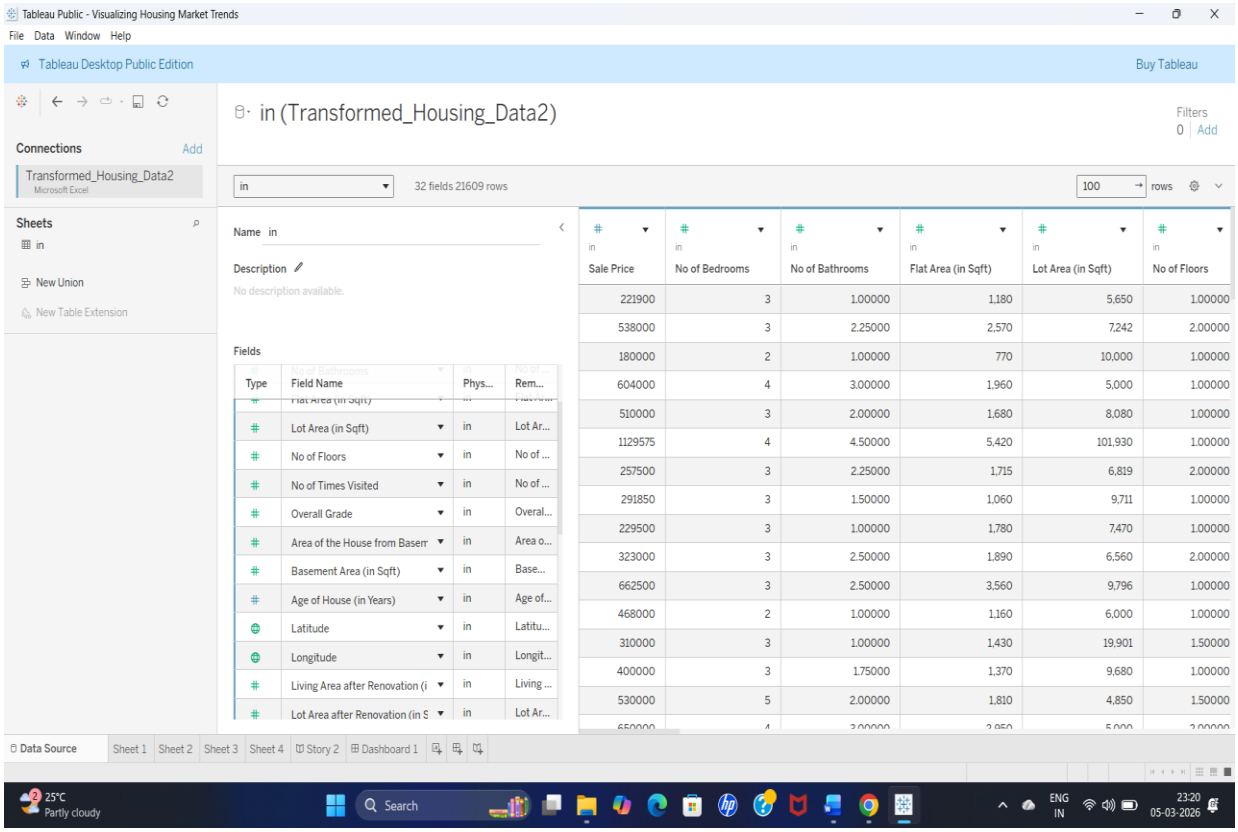
TOOLS AND TECHNOLOGIES USED

Sr. No	Tools	Purpose
1	Tableau	Data Visualization – Used to create interactive dashboards and storyboards for analysing housing market trends.
2	Excel / CSV Dataset	Data Storage – Housing data was collected and stored in Excel/CSV format for further analysis.
3	Spyder (Python IDE)	Data Processing & Analysis – Used to clean, process, and manipulate the dataset using Python before visualization in Tableau.
4	Flask (Python Web Framework)	Web Integration – Used to deploy the dashboards and enable interactive web-based access for users.
5	Web Browser	Dashboard Viewing & Publishing – Tableau dashboards and Flask web apps can be accessed and interacted with online.

DATA COLLECTION & EXTRACTION

The dataset used in this project was collected from the Skill Wallet website and saved in XLSX format. The dataset contains information about housing properties including sale price, number of bedrooms, area, and other features.

The dataset was imported into Tableau using the appropriate connection method. Data fields were verified to ensure correct data types and structure before performing analysis.



DATA PREPARATION

Before visualization, the dataset was carefully prepared to ensure accuracy, consistency, and reliability. Missing values were identified and handled appropriately, either by imputing them with suitable values or by removing incomplete records. Duplicate entries and irrelevant fields were removed to reduce noise and improve clarity.

Data types for each column were verified and converted as necessary, ensuring that numeric fields (such as sale price and house size) and categorical fields (such as location and property type) were correctly defined. Additional features, like price per square foot and property categorization, were derived to enhance analysis. Outliers and anomalies were also detected and addressed to prevent misleading insights.

The data cleaning and preparation process ensured that the dataset was of high quality, consistent, and ready for accurate visualization in Tableau. This step was crucial in producing reliable insights and meaningful interactive dashboards for analysing housing market trends.

DATA VISULIZATION

Various visualization techniques were used to analyse housing market trends. Different charts such as bar charts, line charts, and scatter plots were created in Tableau to represent relationships between housing features and sale prices.

These visualizations help in understanding patterns such as how property size, number of bedrooms, and location influence house prices.

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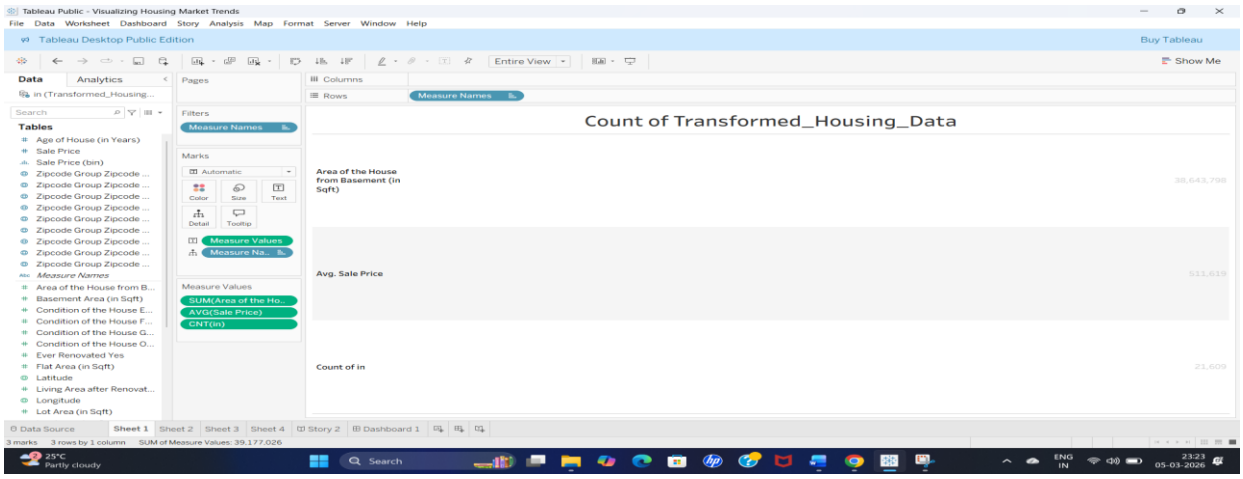

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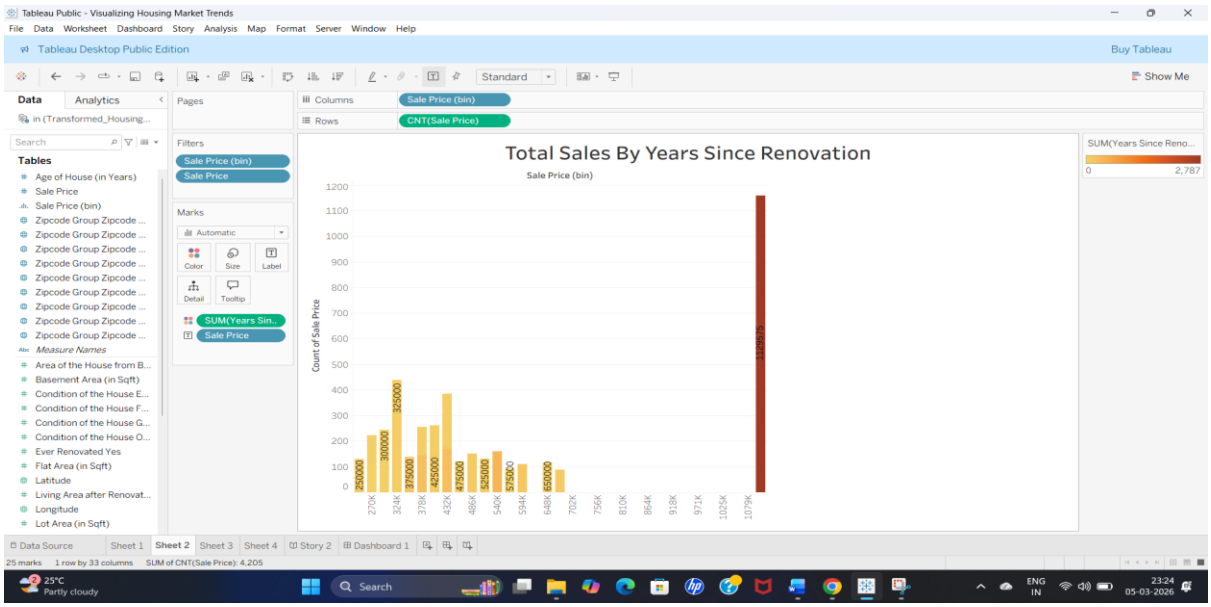

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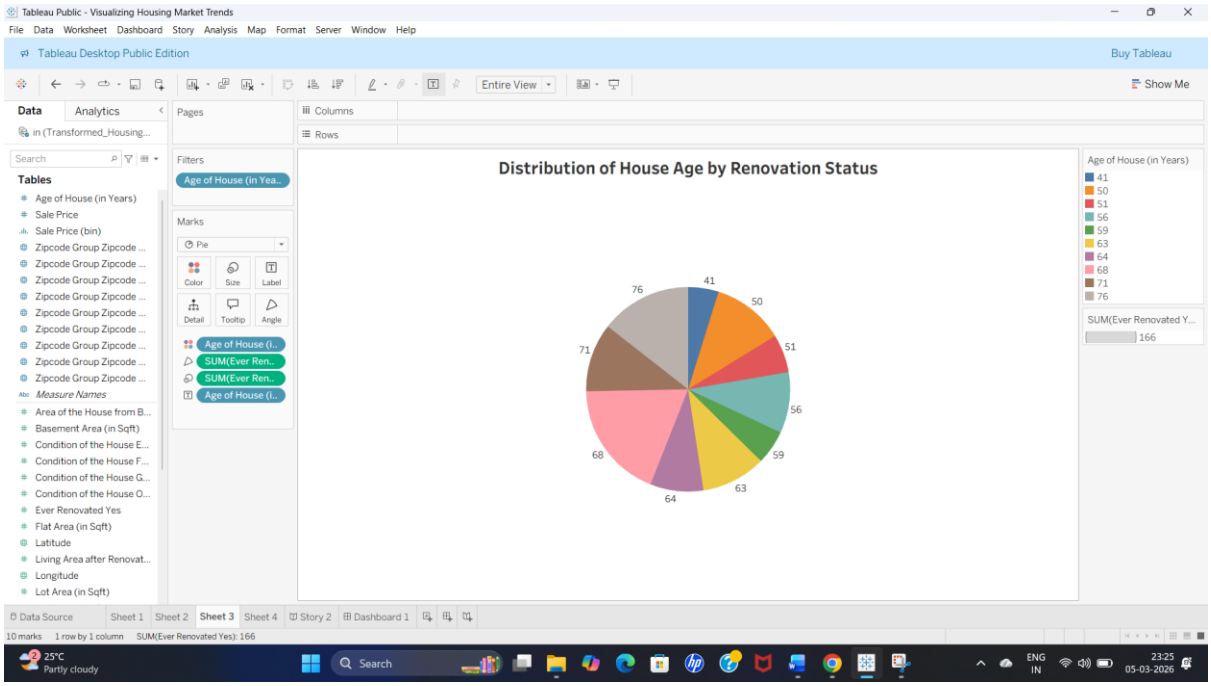

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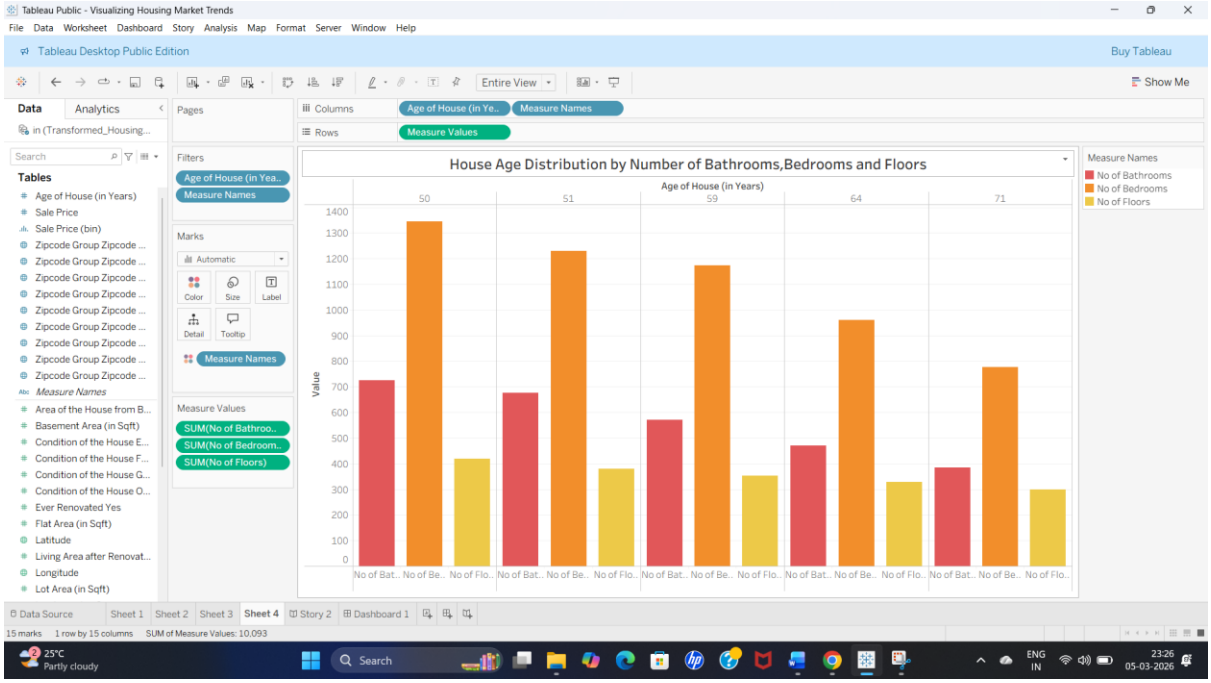
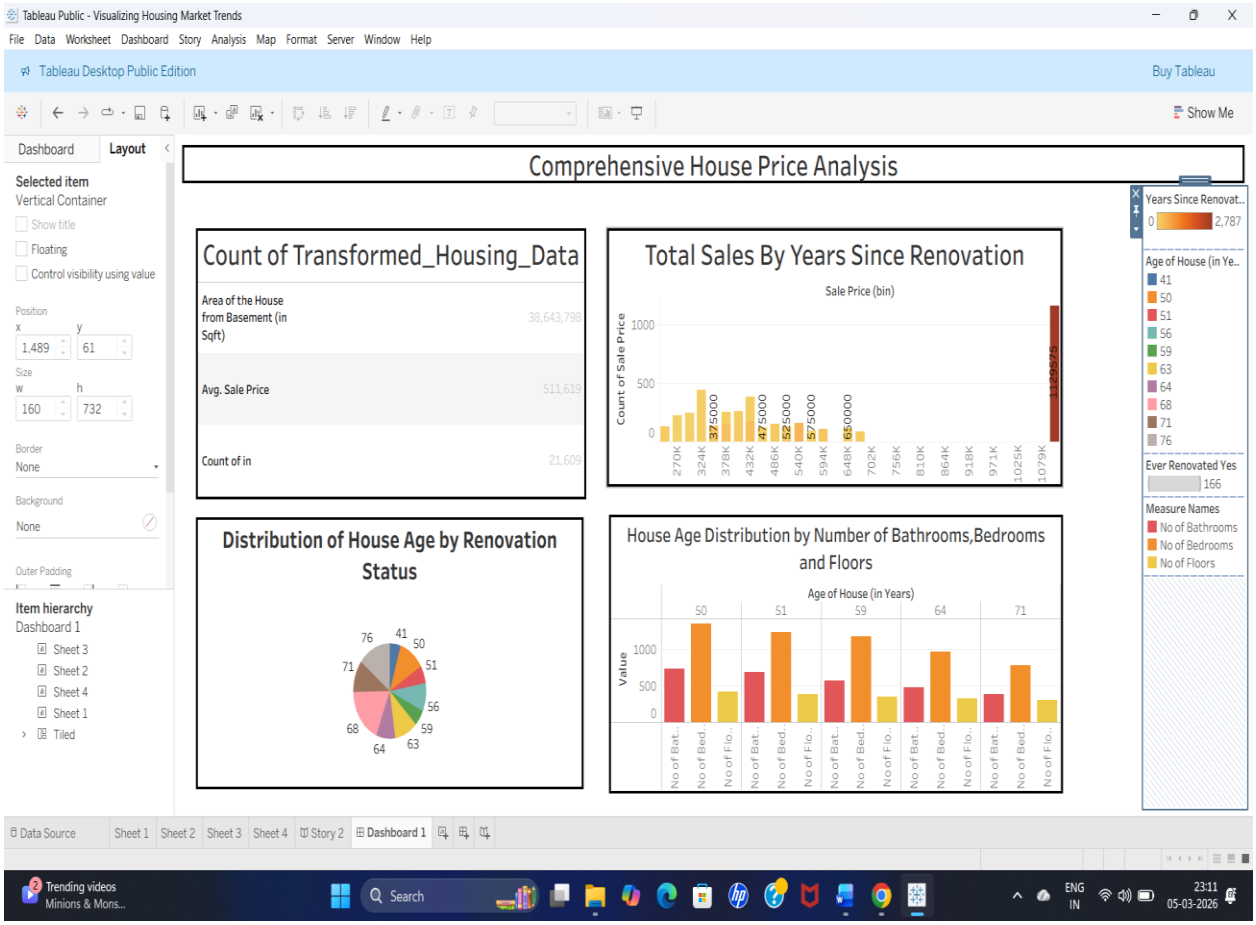


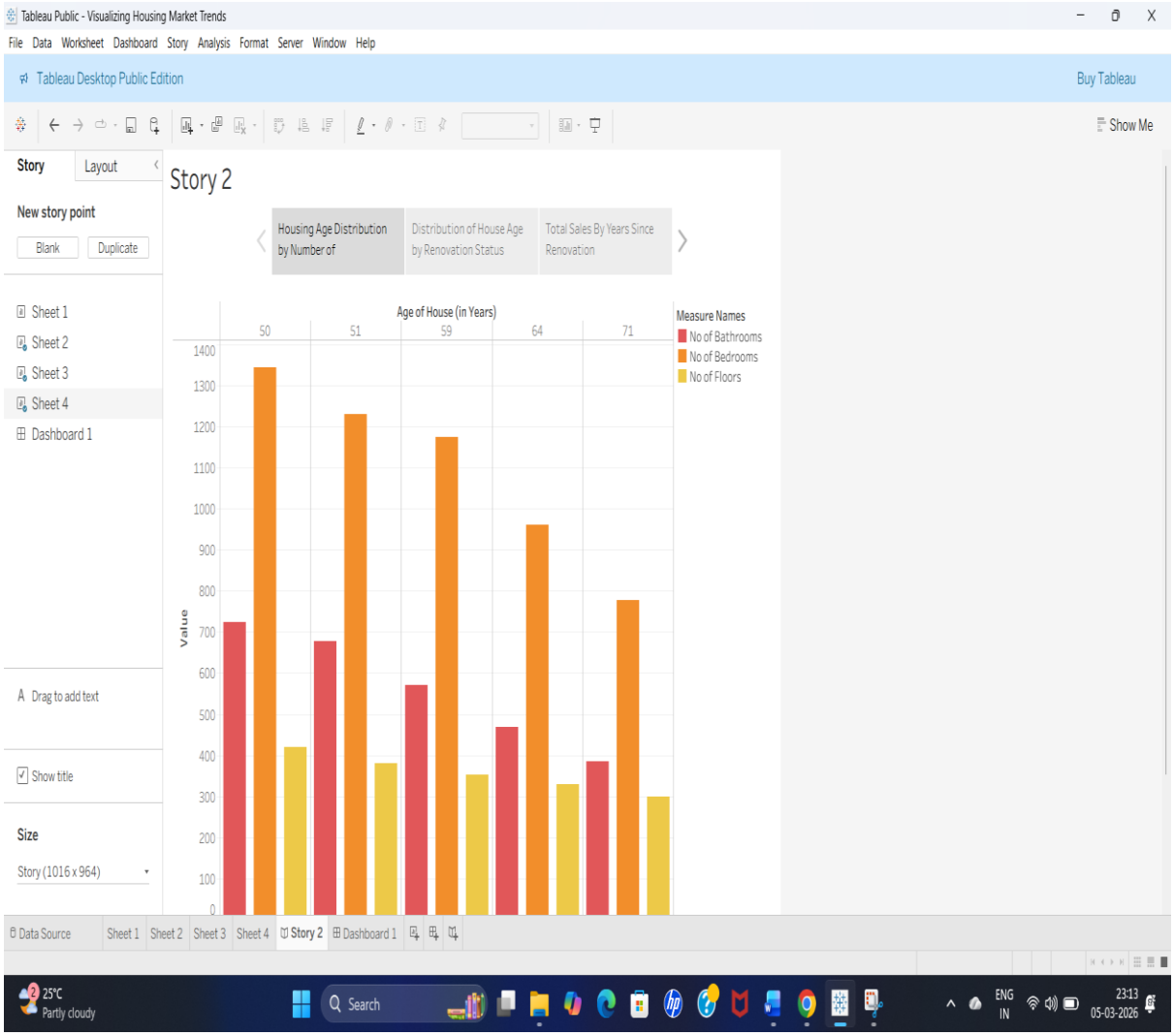
DASHBOARD

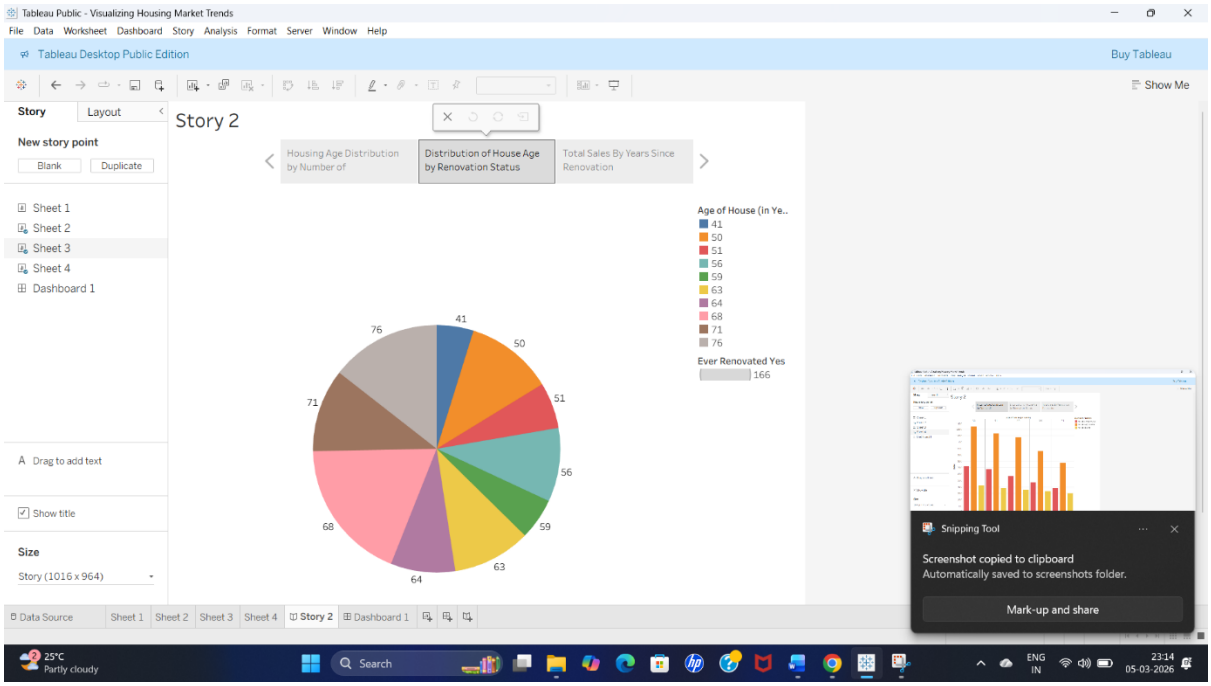
An interactive dashboard was developed in Tableau displaying key insights. It includes multiple visualizations such as price trends, feature impact, and location-based comparisons. Users can filter by area, number of bedrooms, or other features to explore specific market segments. This enables dynamic and actionable insights for stakeholders.



STORYBOARD

A storyboard was created in Tableau to present a structured narrative of the housing market analysis. Each step highlights critical trends, comparisons, and correlations. This guided visualization allows users to understand the market story clearly and make informed decisions based on data insights.

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Sheet 3
Sheet 4
Dashboard 1
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Story 2
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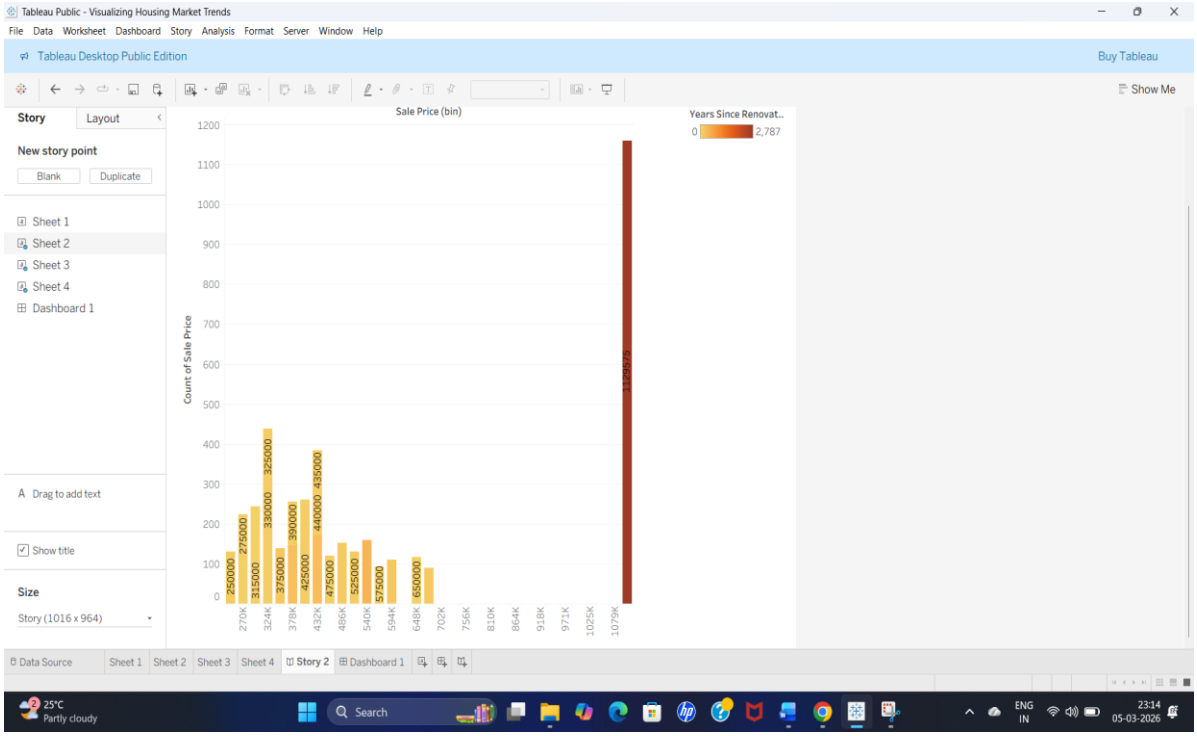
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Sheet 2
Sheet 3
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Story 2
Dashboard 1
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Partly cloudy
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PERFORMANCE TESTING

Performance testing was conducted to ensure that the system and Tableau dashboards functioned efficiently and provided a smooth user experience. The dashboards were tested across different devices, including desktops, laptops, and tablets, to verify that visualizations loaded correctly and remained interactive.

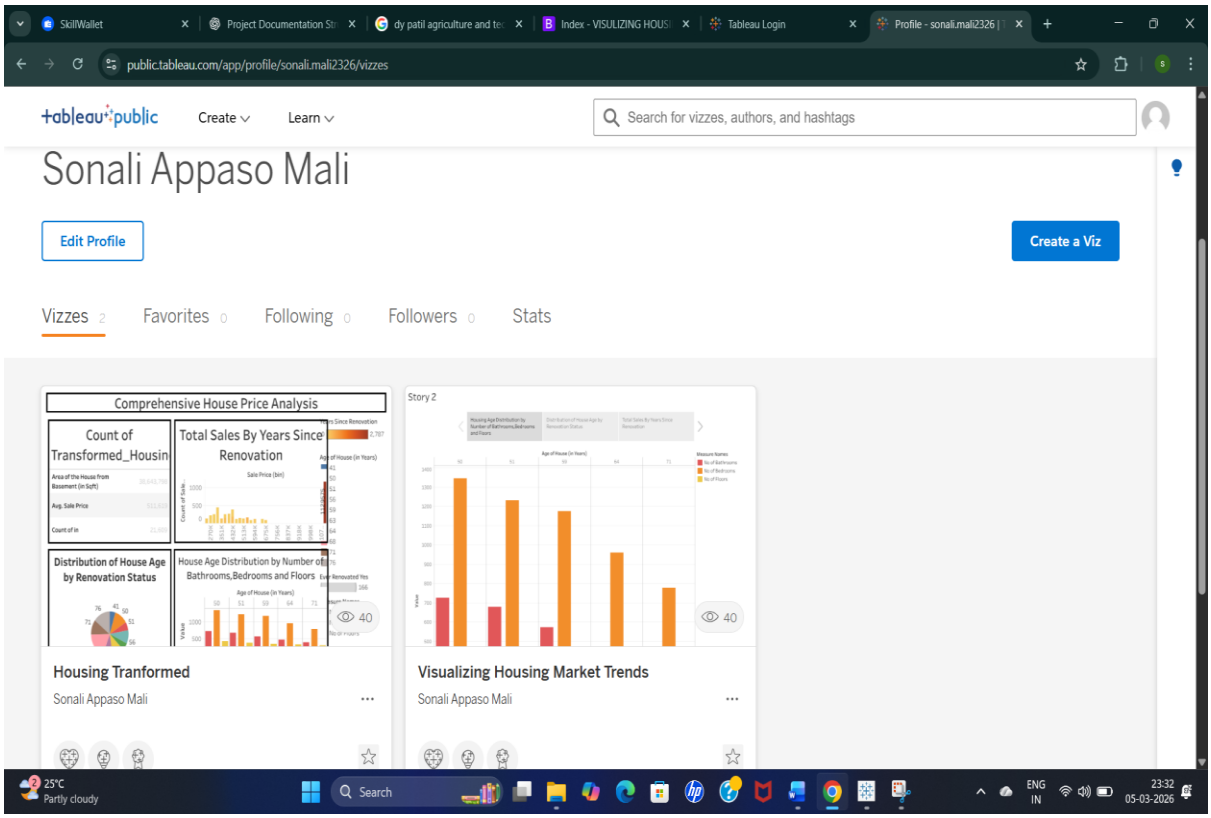
Additionally, multiple web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge were used to check responsiveness, compatibility, and rendering of charts, filters, and dashboards. Interactive elements like drop-downs, sliders, and drill-down options were thoroughly tested to ensure seamless operation without lag or errors.

This testing process ensured that the dashboards and web interface provided consistent, fast, and reliable performance, allowing users to explore housing market trends without technical difficulties. It also confirmed that the system could handle multiple interactions simultaneously, maintaining usability and efficiency for end users.

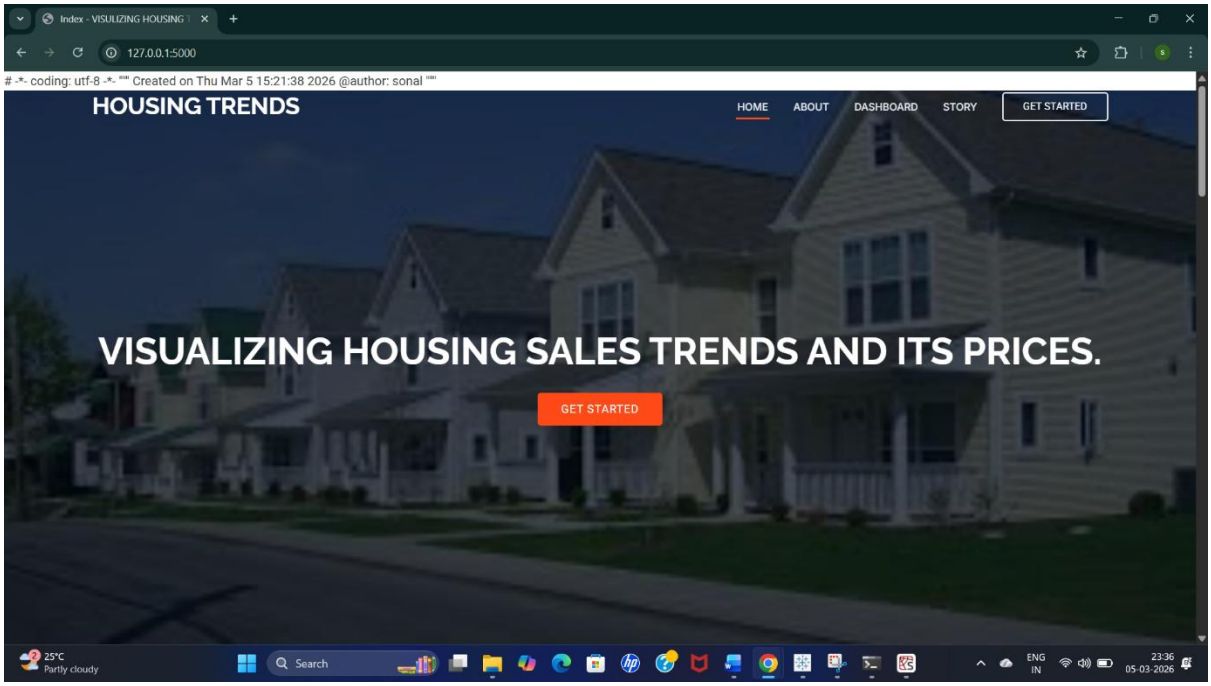
WEB INTEGRATION

The final dashboards were deployed using Flask and Tableau Public. Users can access and interact with the dashboards through a web browser. Flask allowed embedding dashboards and creating a lightweight web interface for dynamic exploration. This ensures accessibility and enhances the practical utility of the project.

Publishing:

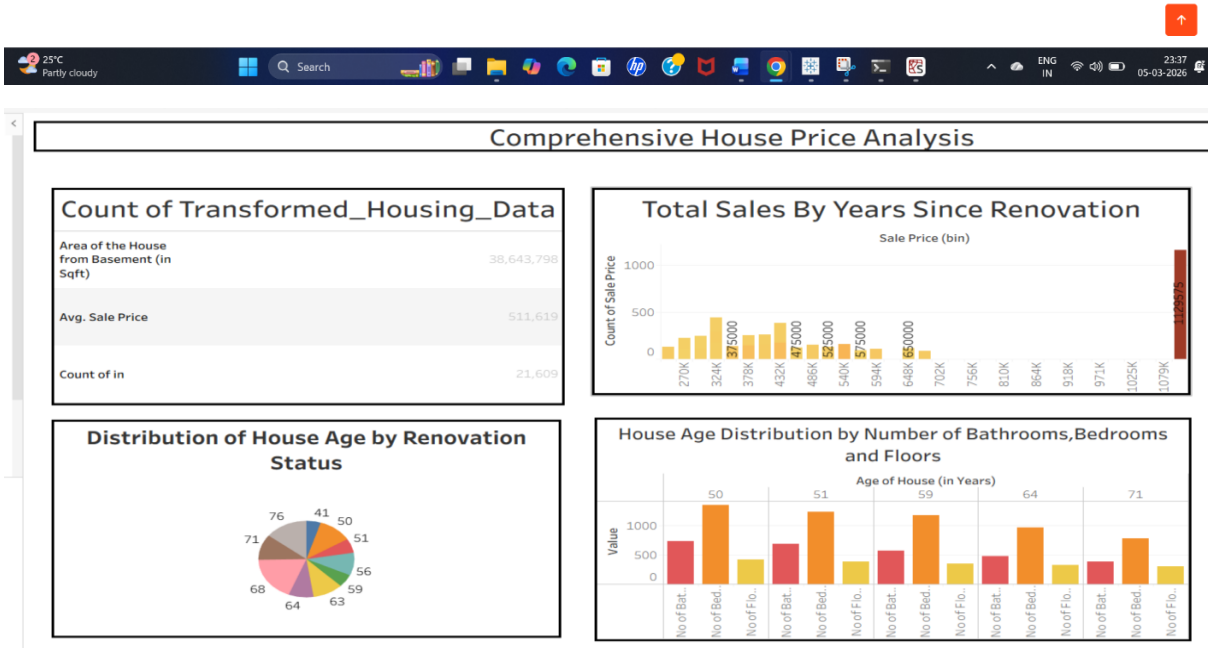


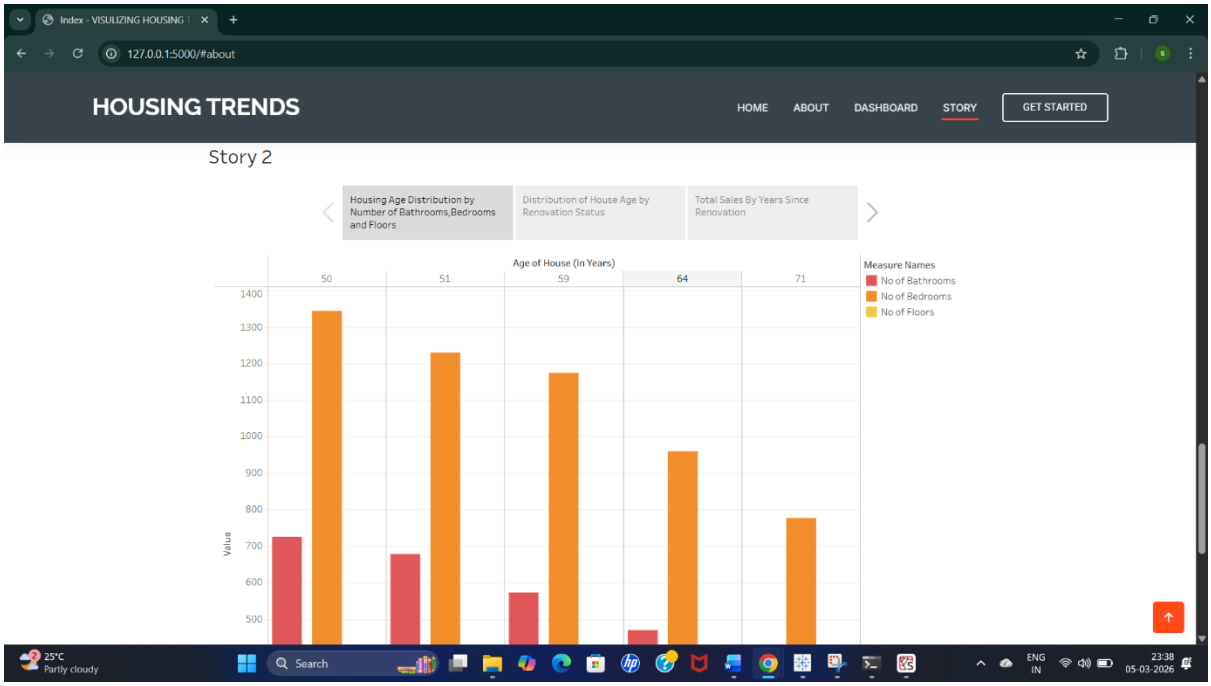
Dashboard and Story embed with UI With Flask:



Housing Market Trends Visualizing: An Analysis of Sale Prices and Features using Tableau

- The goal is to address challenges in understanding the factors that influence house prices and sales trends.
- Using comprehensive housing data, including total sales by years since renovation, house age distribution by bathrooms, bedrooms, and floors.
- Utilizing Tableau to visualize and interpret housing market patterns to support strategic pricing and market competitiveness.

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HOUSING TRENDS
HOME
ABOUT
DASHBOARD
STORY
GET STARTED
Story 2
Housing Age Distribution by Number of Bathrooms,Bedrooms and Floors
Distribution of House Age by Renovation Status
Total Sales By Years Since Renovation
Age of House (in Years)
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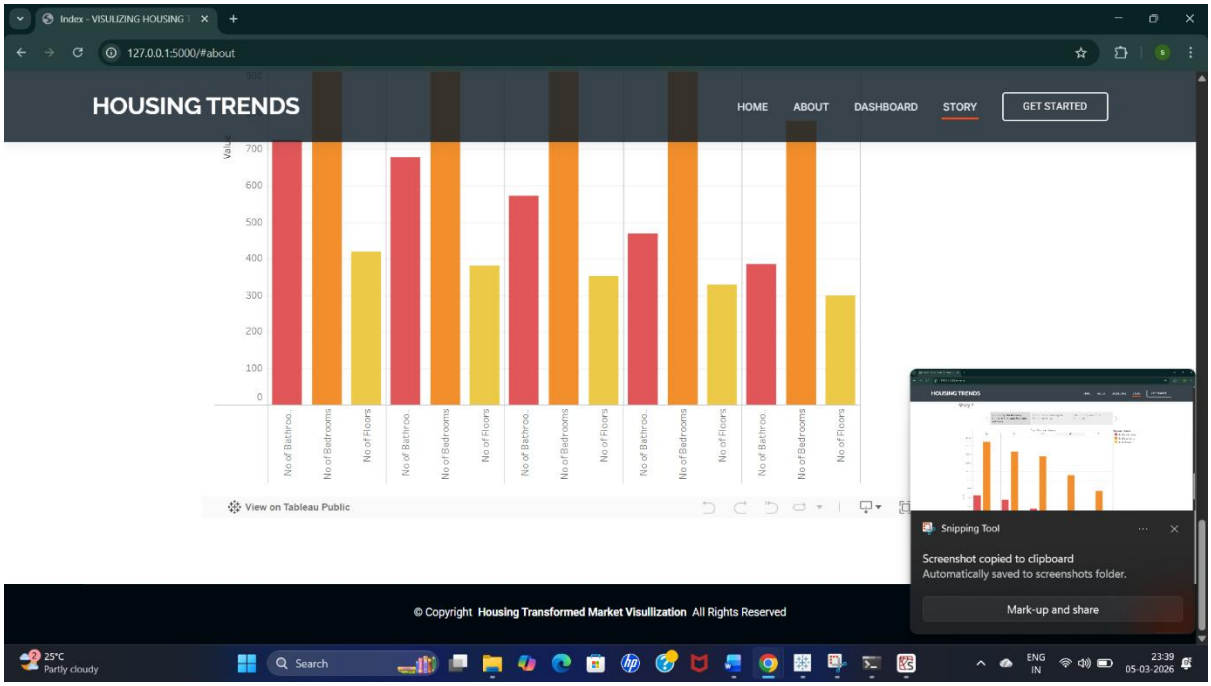
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SYSTEM REQURIMENTS

To successfully run this project and access the interactive dashboards, certain hardware and software requirements must be met. The project relies on **Python, Spyder IDE, Flask, Tableau**, and a web browser for deployment and visualization. Adequate system resources are essential to ensure smooth performance, quick loading of dashboards, and efficient data processing.

Sr. No	Requirement	Details
1	Operating System	Windows 10 / 11 or Linux / macOS
2	Processor	Intel i3 or higher (4th Gen or above)
3	RAM	Minimum 8 GB recommended
4	Storage	Minimum 50 GB free space for datasets and dashboards
5	Software	Python 3.x, Spyder IDE, Flask, Tableau Desktop / Tableau Public
6	Web Browser	Google Chrome / Mozilla Firefox / Microsoft Edge (latest version)
7	Internet Connection	Required for Tableau Public publishing and online dashboard access

CONCLUSION

This project successfully visualized housing market trends using **Tableau** and **Python**, providing a comprehensive understanding of property price patterns. Data-driven analysis identified the key factors influencing house sale prices, including location, size, number of bedrooms, age of the property, and available amenities. Interactive dashboards enabled users to explore these insights in a user-friendly and dynamic manner, making complex data more accessible and understandable.

The integration of **Spyder**, **Tableau**, and **Flask** ensured accurate data processing, effective visualization, and smooth web deployment. Spyder was used to clean, prepare, and analyse the dataset, Tableau transformed the data into meaningful and interactive visualizations, and Flask allowed the dashboards to be deployed on the web for easy accessibility.

Overall, this project demonstrates the power of combining data analytics, visualization, and web technologies to derive actionable insights from housing market data. It provides stakeholders—buyers, sellers, investors, and policymakers—with the tools to make informed decisions and supports a practical approach for real-world applications.

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